Interval arithmetic

x — one number

represented → $x \in [\underline{x}, \overline{x}] = [x]$

$\underline{x}$: lower bound; $\overline{x}$: upper bound.

$x \circ y$ — $\circ$: operation

$\rightarrow$ real result $x \circ y \in [x] \circ [y]$ belongs to resulting interval.

f function

$f(x) \in F(X)$ → composition between a function and an interval

Main objective:

Input $x \mapsto \ldots \rightarrow [a, b]$

Computations done in such a way that we are sure Exact result $\in [a, b]$

A correctly implemented interval arithmetic will give us guaranteed precision on final results.

The final precision of the computation is linked to the size of final interval.

Example: Interval operations ... → [1.01; 1.03]

Error: 0.02

Interval operations ... → [-10; 10]

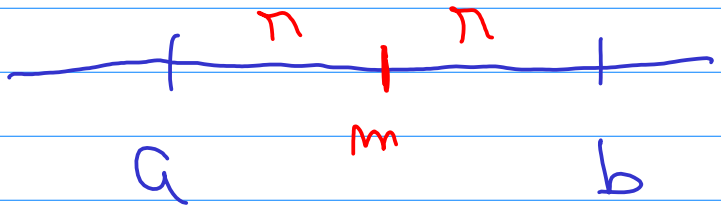
Error: 20

Goal: final interval contains exact result
- size of final interval depends on the implementation / complexity

Many operations → larger intervals

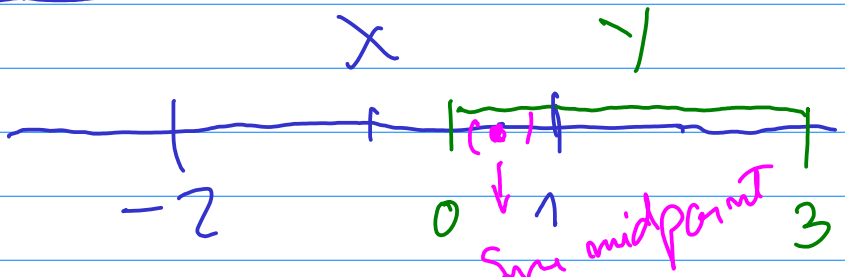
Intervals:

- endpoints $[a, b]$
- midpoint, radius $[m-r, m+r]$



- hull(X, Y) = smallest interval containing X, Y

Question:

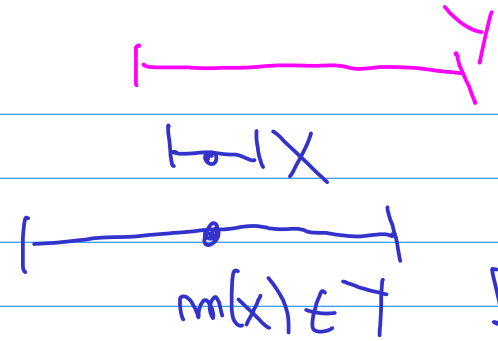


$X \cap Y = [0, 1]$ $\quad w(X) = 3$

$Hull(X, Y) = [-2, 3]$ $\quad w(Y) = 3$

$m(X) = -0.5$ $\quad m(Y) = 1.5$

Is $X \subseteq Y$ (No)



but $X \not\subseteq Y$

midpoint comparison does not decide containment.

Arithmetic operations

$x \circ y \rightarrow$ usual binary operation

Interval operation (set operation)

$$X \circ Y = \{ x \circ y : x \in X, y \in Y \}$$

independent

Addition: $X = [a, b]$, $Y = [c, d]$

$$\left.\begin{array}{l} x \in X, y \in Y \\ a \le x \le b \\ c \le y \le d \end{array}\right\} \Rightarrow a + c \le x + y \le b + d$$

$$x + y \in [a + c, b + d]$$

$X + Y$

Subtraction: $X - Y$?

$$a \le x \le b$$

$$c \le y \le d \quad | \cdot (-1)$$

$$-c \ge -y \ge -d$$

$$-d \le -y \le -c$$

$$\Rightarrow a - d \le x - y \le b - c$$

$$X - Y = [a - d, b - c]$$

$X = [1, 2]$

$Y = [3, 4]$

$$X - Y = [\min X - \max Y ; \max X - \min Y]$$

$$= [-3 ; -1]$$

In general: working with endpoints is not enough.

working with midpoints is not enough.

Addition: $X = [m - r, m + r]$

$Y = [n - s, n + s]$

$$X + Y = [(m+n) - (r+s), (m+n) + (r+s)]$$

$$r(X+Y) = r(X) + r(Y)$$

→ repeating additions increases size of intervals

Subtraction: $X = [m - r ; m + r]$

$Y = [n - s ; n + s]$

$$X - Y = [m - r - n - s ; m + r - n + s]$$

$$= [m - m - (r+s) ; m - n + (r+s)]$$

→ subtract midpoints | Radius = $\frac{\text{width}}{2}$

→ ADD the radii

Multiplication

$X = [a, b]$

$Y = [c, d]$

$$XY = [\min\{ac, ad, bc, bd\}; \max\{ac, ad, bc, bd\}]$$

- $X = [1, 2]$, $Y = [5, 9]$, combinations $\{5, 9, 10, 18\}$

  $XY = [5, 18]$

- $X = [-2; 3]$, $Y = [-4; -1]$

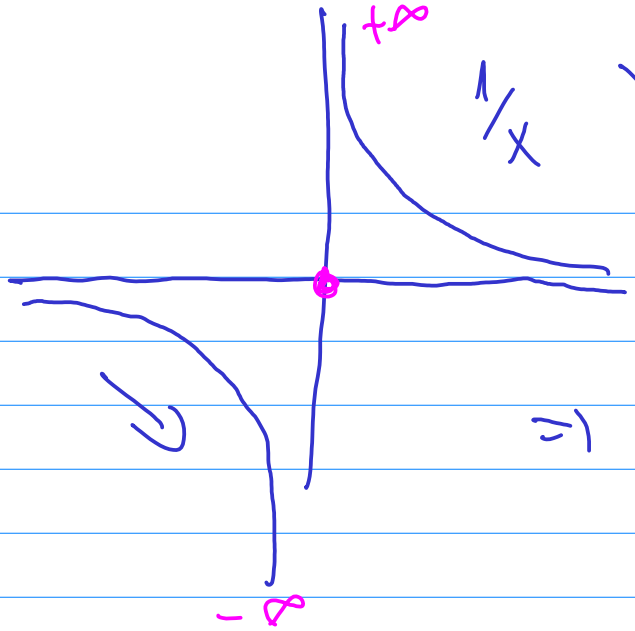
  combinations: $\{8; 2; -12, -3\}$

  $\Rightarrow XY = [-12; 8]$

Division? Multiplication with the inverse

$$\frac{x}{y} = x \cdot \frac{1}{y}$$ well defined for $y \neq 0$

If we want to divide by $Y = [c, d]$ then we assume $0 \notin Y$.

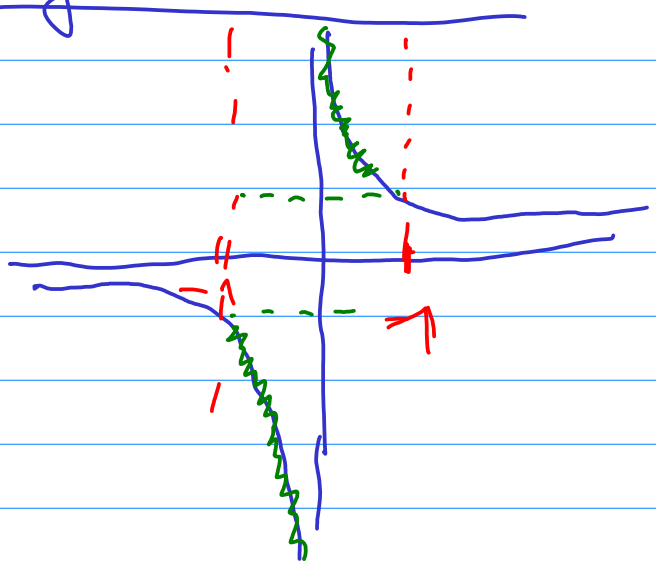


$$\Rightarrow \frac{1}{Y} = \left[\frac{1}{d}; \frac{1}{c}\right]$$

(same for positive/negative)

Define $\frac{X}{Y} = X \cdot \left(\frac{1}{Y}\right)$

Why exclude zeros



$$\frac{1}{[-1,1]} = (-\infty; -1] \cup [1, \infty) \cup \{\pm\infty\}$$

↓

Not an interval

Computing the square

X^2, $X = [a, b]$

Ideal: $X^2 = X \cdot X$

$X = [-2;3]$

endpoint op: $\{4, -6, -6, 9\}$

$[-2;3] \cdot [-2;3] = [-6;9]$

correct.

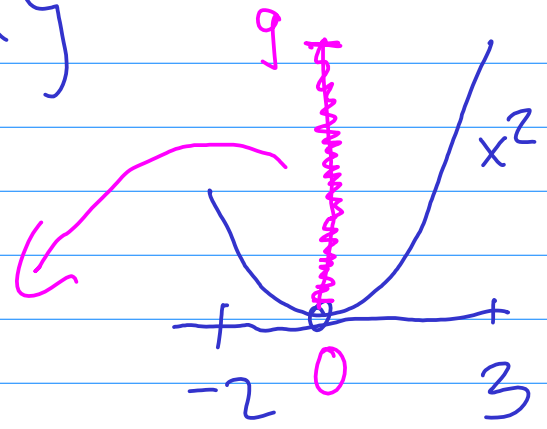
$X \cdot X$ contains all possible squares.

Not optimal (tight enough)

$$X^2 = \{ x^2 : x \in X \}$$

Remember: $x^2 \geq 0$

$\Rightarrow X^2 = [0;9]$



Rule

$$\text{square}([a;b]) = \begin{cases} [a^2, b^2] & 0 \leq a \\ [b^2, a^2] & b \leq 0 \\ [0; \max(a^2, b^2)] & a < 0 < b \end{cases}$$

Parallel: $X - X$

$X = \{x\} \Rightarrow X - X = \{0\}$

$X = [a;b] \rightarrow X - X = [a-b; b-a]$

Computation:

- if we have correctly defined math interval operations
- if operations are well defined

Finite set of admissible operations

→ enclose the desired result.

Well def: - no division by 0
- no negatives inside square root
- no negatives or 0 inside a log

...

Working with intervals (correctly) can produce math proofs.

$$X = [2,3] \quad , \frac{1}{X} = \left[\frac{1}{3}; \frac{1}{2}\right]$$

$$X + \frac{1}{X} = \ldots = \left[\frac{7}{3}; \frac{7}{2}\right]$$

$$\Rightarrow \quad X + \frac{1}{X} > 0 \quad \forall X \in [2,3]$$

$$x + \frac{1}{x} \geq \frac{7}{3} \quad \forall x \in [2;3]$$

→ math proof obtained through interval computations

Careful with algebra

$$X(Y+Z) \subseteq XY + XZ$$

$X = [-1,1]$, $Y = [1,1]$, $Z = [-1,-1]$

$Y + Z = [0,0]$

$\Rightarrow X \cdot (Y+Z) = [0,0]$

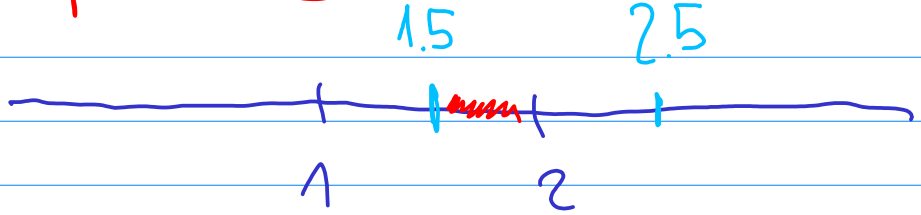
$XY + YZ = [-2;2]$ ↔ different

Exercise 3

Program 1 ⟼ Result $\in [1,2]$

Program 2 ⟼ Result $\in [\frac{3}{2}; \frac{5}{2}]$

Improved inclusion:



If both programs are correct

⇒ Result ∈ intersection

$\in \left[\frac{3}{2}; 2\right]$ ✓

Ex 4 $\quad x + \frac{1}{x}$, $X = [2, 3]$

$$X + \frac{1}{X} = \left[\frac{7}{3}; \frac{7}{2}\right]$$

image of $x + \frac{1}{x}$ on $[2, 3]$

$$f'(x) = 1 - \frac{1}{x^2} > 0$$

function ↗

$$\Rightarrow \text{Range } f(x) = [f(2); f(3)]$$

$$= \left[\frac{5}{2}; \frac{10}{3}\right]$$

↓

tight or bounds

than $X + \frac{1}{X}$

$X = [2, 3]$

$Y = [1, 2]$

$X \cdot Y = [2, 6]$ tight enclosure

$[0, 9]$ is an enclosure of $X \cdot Y$

because it contains every possible result in xy with $x \in X, y \in Y$